# The Chronos Protocol: Computational Time-Locks

## 6.1 The Fallacy of Centralized Custody

Traditional digital time capsules rely on a "Trusted Third Party" to store keys and release them at a specific date. This introduces a **fatal architectural weakness**: if the service provider ceases to exist, the data becomes permanently inaccessible. To achieve true digital survivability, a system must decouple the "Time-Lock" from the "Server's existence."

## 6.2 The Concept: Nonce as a Source of Time

In the PSQC (Phantom-Step Cascade Container - The Chronos Protocol) model, time is not measured by a calendar, but by computational work. The system utilizes a random nonce as the secondary key. By withholding a portion of this nonce, we transform a cryptographic secret into a "Time-Lock."

The difficulty of unlocking is defined by the search space of the missing bits:

$$T \approx \frac{2^{bits}}{V_{hash}}$$

Where $V_{hash}$ is the velocity of verification attempts per second.

## 6.3 Architecture of the PSQC Container

A PSQC container encapsulates the encrypted payload with specific metadata required for both server-assisted and offline recovery:

- **nonce_hint:** A masked representation of the nonce (e.g., `hF**********` ). The number of masked characters determines the "computational depth" of the time-lock.
- **nonce_hash_V1:** The SHA-256 target. The only way to unlock the container is to find a candidate that satisfies: `Hash(Candidate_Nonce) == Target_Hash` .
- **ciphertext:** The payload, encrypted using the derived key.

Unlike traditional archives, a `.PSQ` file is a stateless entity that functions as a computational time-lock:

```
{
  "protocol": "Phantom-Step/V4",
  "nonce_hint": "LEdWTvvka9******",
  "nonce_hash_V1": "48d49beb8cbb82c65...4f3b928",
  "ciphertext": "base64_payload_data...",
  "integrity_check": "AEAD_Poly1305_Tag"
}
```

## 6.4 Hybrid Recovery Paths

The Chronos Protocol implements a dual-layered recovery mechanism:

- **The Fast Path (Server-Assisted):** An optional server stores the full nonce and releases it only after a specified timestamp. This allows for near-instant decryption upon maturity.
- **The Autonomous Path (Offline Discovery):** If the server is offline, the user can "brute-force" the remaining bits. The `nonce_hint` allows calibration of the search space from minutes to decades.

## 6.5 Memory-Hard Time Enforcement

To prevent high-speed parallel attacks (GPU/ASIC farms), the PSQC model integrates **Argon2id** as a mandatory verification step for every nonce candidate.

> By requiring significant RAM (e.g., 1GB) for each attempt, the "time-lock" becomes resistant to hardware scaling. Time is effectively measured in memory cycles, ensuring a user with a home PC can reasonably compete with a centralized entity.

## 6.6 Security Properties & Independence

- **Zero Server-Side Secrets:** The server never stores the ciphertext or the user's master key. It only holds the nonce.
- **Deterministic Integrity:** Once found, the nonce is fused with the physical anchor to generate the final key.
- **Portability:** The PSQC container is self-contained and can be stored on any cold storage or public cloud.

## Experimental Prototype

Implemented in **SecretMemoryLocker v4.05+**. Test the integration of server-side gating and client-side computational discovery.